

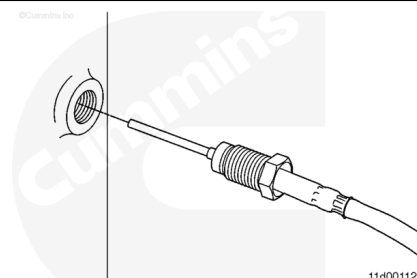
Trainings ▾

Initial Check

For CM871 and CM876 engine remove the aftertreatment exhaust gas temperature sensor from the aftertreatment inlet and disconnect it from the wiring harness. Install the pressure gauge adapter, Part Number 4918576. Connect a manometer or pressure gauge, Part Number ST-1273, to the pressure gauge adapter.

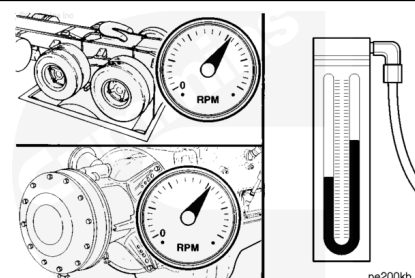
Protect the hose from heat. Use a 305 mm [12 in] minimum length of metal tubing leading from the exhaust pipe connection.

Note : Disconnecting the aftertreatment DOC inlet temperature sensor will trigger Fault Code 1666. Do not troubleshoot this fault code during this exhaust restriction measurement.



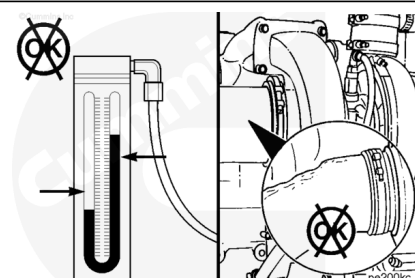
Operate the engine at rated rpm and load. Record the reading on the manometer.

Exhaust Restriction				
		mm-hg	in-hg	in-h ₂ O
Without Aftertreatment	MAX	75.0	3.0	40.8
With Aftertreatment	MAX	305.0	12.0	163.3



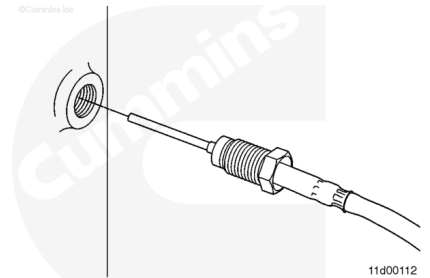
If the exhaust pressure exceeds the specifications, inspect the exhaust piping for damage. Refer to the OEM troubleshooting and repair manual.

If the vehicle is equipped with an aftertreatment system, inspect the aftertreatment system for a plugged aftertreatment DPF and/or a plugged aftertreatment DOC.



Extended engine operation with high exhaust restriction can lead to damaged turbocharger oil seals. Use the following procedure to inspect the turbocharger for progressive damage. Refer to Procedure 010-033 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html>)

Remove the test equipment and install the aftertreatment DOC inlet temperature sensor. Operate the engine and verify that all fault codes are inactive.



Preparatory Steps

- Perform a stationary regeneration, if possible, to make sure the aftertreatment is clean prior to performing the exhaust restriction test. Refer to Procedure 014-013 in Section 14. (</qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html>)

Note : Some fault codes may prevent a stationary regeneration from being started/completed. If that is the case, move on to the next step.



Measure

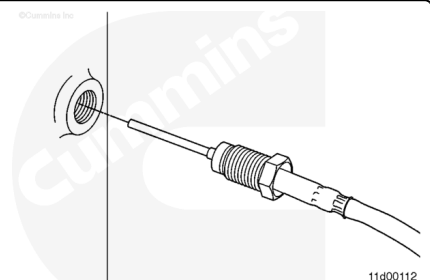
Remove the aftertreatment exhaust gas temperature sensor. Refer to Procedure 019-449 in Section 19.

(</qs3/pubsys2/xml/en/procedures/181/181-019-449.html>)

Install the pressure gauge adapter, Part Number 4918576.

Connect a manometer or pressure gauge, Part Number ST-1273, to the pressure gauge adapter.

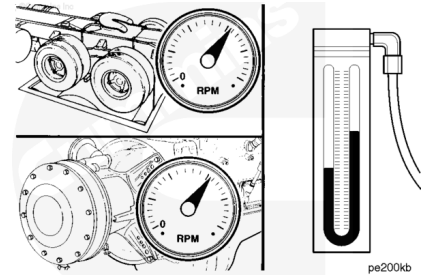
Protect the hose from heat. Use a 305 mm [12 in] minimum length of metal tubing leading from the exhaust pipe connection.



Note : Disconnecting the aftertreatment DOC inlet temperature sensor will trigger Fault Code 3314. Do not troubleshoot this fault code during this exhaust restriction procedure.

Note : To achieve an accurate exhaust restriction test, be sure a stationary regeneration was performed prior to conducting the exhaust restriction test.

Operate the engine at rated speed and load and record the exhaust restriction. Refer to Procedure 018-020 in Section V. (</qs3/pubsys2/xml/en/procedures/181/181-018-020.html>)



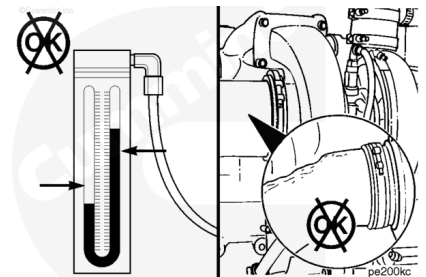
If the exhaust restriction exceeds the specifications, inspect the exhaust piping for damage. Refer to the OEM service manual.

If the exhaust restriction exceeds the specifications:

1. Check for a plugged aftertreatment diesel particulate filter (DPF).
2. Check for a plugged DOC.
3. Check for a damaged or restricted selective catalytic reduction (SCR) decomposition tube and SCR catalyst.

Extended engine operation with high exhaust restriction can lead to damaged turbocharger oil seals. Use the following procedure to inspect the turbocharger for progressive damage. Refer to Procedure 010-033 in Section 10.

(</qs3/pubsys2/xml/en/procedures/181/181-010-033-tr.html>)



Finishing Steps

- Remove the test equipment and install the aftertreatment exhaust gas temperature sensor. Refer to Procedure 019-449 in Section 19. (</qs3/pubsys2/xml/en/procedures/181/181-019-449.html>)
- Operate the engine and verify that all fault codes are inactive.



